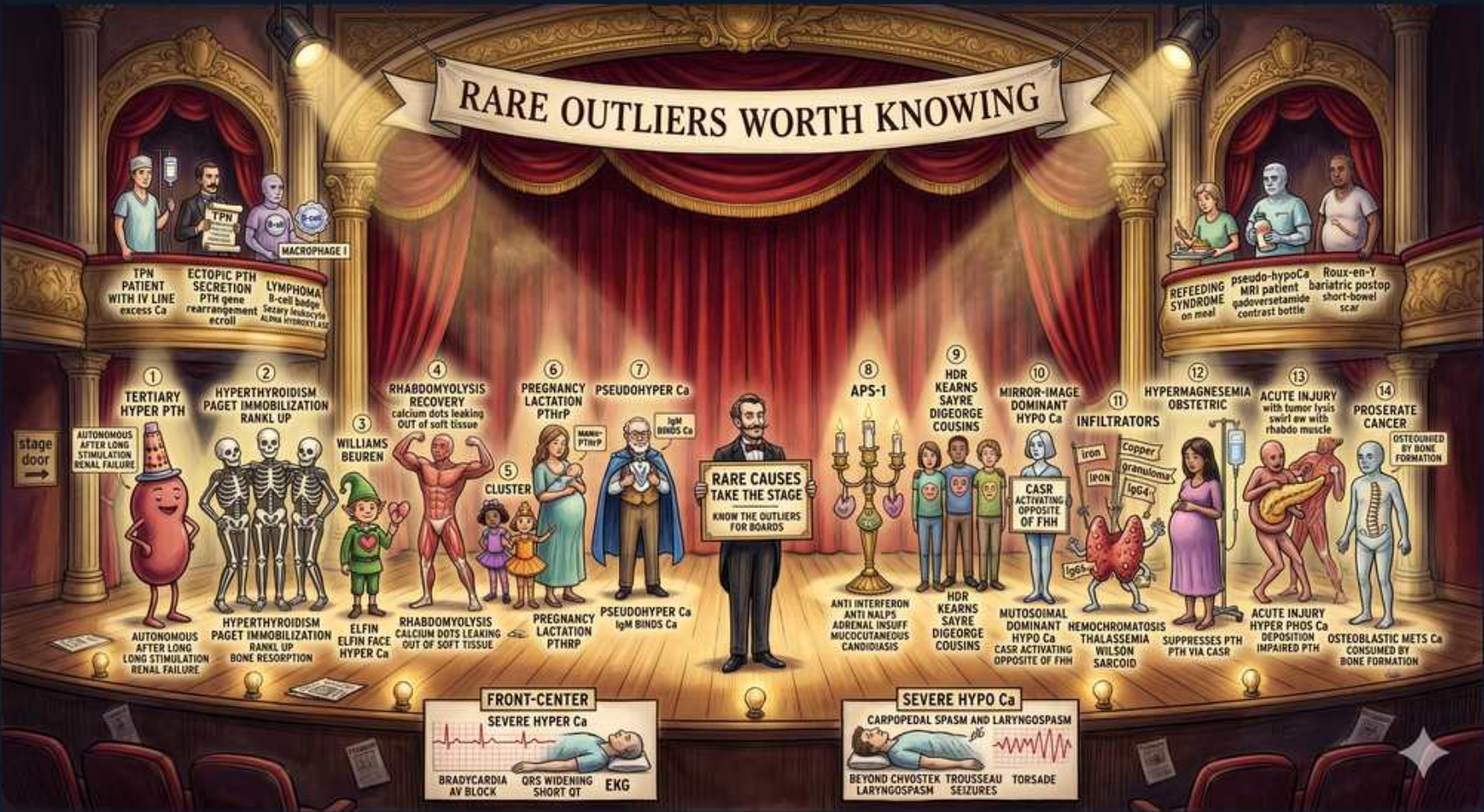


Hypercalcemia & Hypocalcemia — Rare Outlier Causes for Boards



1. PTHrP — humoral hypercalcemia

WHAT YOU SEE

Figure secreting PTHrP into a line

WHAT IT ENCODES

PTHrP from squamous cell lung, breast, renal cell, and head/neck cancers drives humoral hypercalcemia of malignancy. PTHrP mimics PTH at the receptor: high Ca, LOW phosphate, but PTH itself is suppressed and 1,25-vitD is low. This is the most common mechanism of malignancy-associated hypercalcemia.

BOARD TRAP

Low PTH with high Ca looks like the tumor 'makes PTH' — but ectopic true PTH is vanishingly rare; PTHrP (a separate assay) is the answer.

2. Ectopic PTH secretion

WHAT YOU SEE

Tumor labeled ectopic PTH

WHAT IT ENCODES

Genuine ectopic secretion of intact PTH (not PTHrP) is a true rarity, reported in ovarian, lung, and neuroendocrine tumors. Here the standard PTH assay is inappropriately elevated despite hypercalcemia, mimicking primary hyperparathyroidism.

BOARD TRAP

Don't reflexively call high-PTH hypercalcemia 'primary hyperparathyroidism' — imaging-occult ectopic PTH tumor is the rare alternative.

3. Tertiary hyperparathyroidism

WHAT YOU SEE

Red kidney figure with party hat

WHAT IT ENCODES

Long-standing secondary hyperparathyroidism in CKD/after transplant becomes autonomous: parathyroid glands hypertrophy and keep oversecreting PTH even after the renal stimulus resolves, producing HIGH Ca + HIGH PTH. Treat with cinacalcet or parathyroidectomy.

BOARD TRAP

Secondary hyperPTH has LOW/normal Ca; tertiary is distinguished by autonomous hypercalcemia after renal correction/transplant.

4. Immobilization hypercalcemia

WHAT YOU SEE

Skeleton in bed / Paget

WHAT IT ENCODES

Prolonged immobilization with high baseline bone turnover (adolescents, Paget disease, spinal cord injury) uncouples resorption from formation, releasing skeletal Ca. PTH is suppressed; treat with bisphosphonates and mobilization.

BOARD TRAP

Suppressed PTH + immobilized young patient points to bone-resorption hypercalcemia, not malignancy.

5. Williams syndrome

WHAT YOU SEE

Elfin-featured child figure

WHAT IT ENCODES

Williams syndrome (7q11.23 elastin deletion) causes idiopathic infantile hypercalcemia with elfin facies, supravalvular aortic stenosis, and a hypersocial 'cocktail party' personality. Hypercalcemia is from abnormal vitamin D sensitivity.

BOARD TRAP

Elfin facies + hypercalcemia + supravalvular AS = Williams, not DiGeorge (which causes HYPOcalcemia).

6. Rhabdomyolysis recovery phase

WHAT YOU SEE

Muscle breakdown figure

WHAT IT ENCODES

Acute rhabdomyolysis causes EARLY hypocalcemia (Ca deposits in damaged muscle). In the RECOVERY/diuretic phase, that calcium is remobilized causing rebound HYPERcalcemia, often with rising vitamin D.

BOARD TRAP

Rhabdo is classically hypocalcemic — the rebound hypercalcemia appears only during recovery, a frequently tested reversal.

7. Pregnancy / lactation PTHrP

WHAT YOU SEE

Pregnant/lactating figure

WHAT IT ENCODES

Breast tissue and placenta can produce PTHrP; massive mammary hyperplasia or lactation rarely cause hypercalcemia. PTH is suppressed, PTHrP elevated. Resolves with delivery or weaning.

BOARD TRAP

Hypercalcemia in lactation isn't always malignancy — benign PTHrP from breast tissue is the rare physiologic cause.

8. Pseudohypercalcemia

WHAT YOU SEE

Lab tube labeled pseudo Ca

WHAT IT ENCODES

Severe hyperalbuminemia (dehydration) or paraproteins (myeloma) raise TOTAL calcium while IONIZED calcium is normal — a spurious result. Always check ionized Ca or correct for albumin before treating.

BOARD TRAP

Treating a high total Ca without checking ionized Ca/albumin risks iatrogenic hypocalcemia when the elevation was artefactual.

9. Adrenal insufficiency (Addisonian)

WHAT YOU SEE

AIS / Addison figure

WHAT IT ENCODES

Cortisol deficiency causes mild hypercalcemia via reduced renal Ca excretion and hemoconcentration plus increased bone resorption. Resolves rapidly with glucocorticoid replacement.

BOARD TRAP

Hypercalcemia + fatigue + hyponatremia + hyperkalemia should prompt an adrenal-insufficiency work-up, not just a malignancy hunt.

10. Familial hypocalciuric hypercalcemia (FHH)

WHAT YOU SEE

CASR receptor figure

WHAT IT ENCODES

Inactivating CaSR mutation raises the set-point so PTH stays mildly high and the kidney reabsorbs Ca avidly: mild hypercalcemia, mild high PTH, and a LOW urine Ca/creatinine clearance ratio (<0.01). Benign, autosomal dominant, needs NO parathyroidectomy.

BOARD TRAP

FHH mimics primary hyperPTH — the 24-h urine calcium (low in FHH, high in primary hyperPTH) is the discriminator that prevents needless surgery.

11. Granulomatous / sarcoid 1-alpha-hydroxylase

WHAT YOU SEE

Granuloma macrophage figure

WHAT IT ENCODES

Sarcoid, TB, and other granulomas express extrarenal 1-alpha-hydroxylase, converting 25-vitD to active 1,25-vitD autonomously. Result: hypercalcemia with HIGH 1,25-vitD, LOW PTH, often worsened by sunlight. Treat with steroids.

BOARD TRAP

Granulomatous hypercalcemia is driven by 1,25-vitD (active), not 25-vitD — and responds to glucocorticoids, unlike PTH-driven disease.

12. Milk-alkali / thiazide

WHAT YOU SEE

Antacid + thiazide tablets

WHAT IT ENCODES

Milk-alkali (calcium carbonate + alkali) gives hypercalcemia, metabolic alkalosis, and AKI. Thiazides reduce urinary Ca excretion, unmasking or worsening hypercalcemia. Both are reversible by stopping the agent.

BOARD TRAP

A patient on thiazide + calcium supplements with hypercalcemia may not have an endocrine tumor — stop the drugs and recheck first.

13. Carpopedal spasm — severe hypocalcemia

WHAT YOU SEE

Bottom banner: spasm/laryngospasm

WHAT IT ENCODES

Acute severe hypocalcemia produces neuromuscular irritability: Chvostek and Trousseau signs, carpopedal spasm, laryngospasm, prolonged QT, and seizures. Emergency treatment is IV calcium gluconate.

BOARD TRAP

Prolonged QT and tetany in a post-thyroidectomy patient = hypocalcemia, not a primary arrhythmia — give IV calcium, not antiarrhythmics.

14. Bone-image dominant infiltration

WHAT YOU SEE

Marrow-infiltration figure

WHAT IT ENCODES

Osteolytic metastases and myeloma release Ca by direct bone destruction (local osteolytic hypercalcemia), with suppressed PTH and high bone-resorption markers; bisphosphonates/denosumab are key.

BOARD TRAP

Local osteolytic hypercalcemia (direct lysis) is mechanistically distinct from PTHrP humoral hypercalcemia — both occur in cancer but PTHrP gives low phosphate.

15. Acute kidney injury phases

WHAT YOU SEE

Injured kidney figure

WHAT IT ENCODES

AKI alters calcium handling: oliguric phase tends toward hypocalcemia (phosphate retention), while recovery/diuretic phases and rhabdo-AKI can flip to hypercalcemia as deposits remobilize.

BOARD TRAP

Calcium direction in AKI depends on the phase — don't assume a single trend across the illness course.